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Instruction

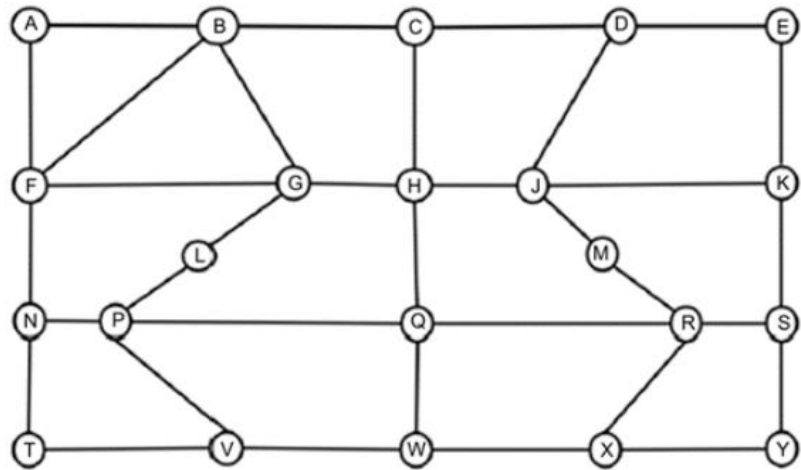
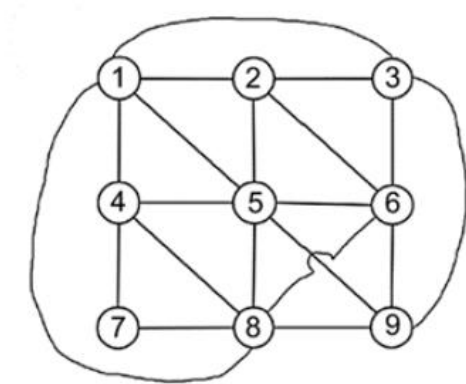
What the question is

Exercises:

Computer Science Problems Worksheet:

In these graphs, find:

- a) the largest clique
- b) the largest independent set

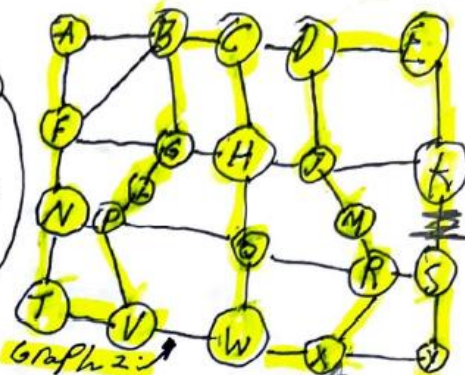
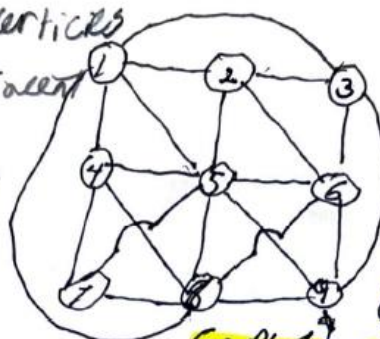


Computer science prob worksheet. VERTS = vertices

In these graphs find: 'Adj = adjacent'

a) The largest clique

b) The largest Independent set



Graph 1:

clique

A) Clique - graph subset of vert such that every 2 distinct vertices = adj;
Largest clique = max # of vertices.
 $\{1, 2, 3, 4, 5, 6\}$ → 5 vertices (just from eyeballing it)

Graph 2:

$\{B, C, G, H, L, P\}$ → 6 vertices

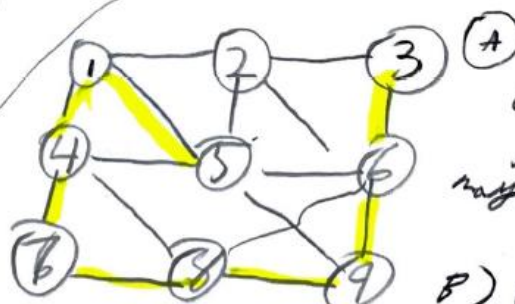
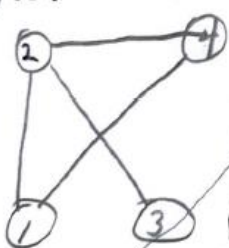
B) Independent set set of vert such that 2 vertices = adj.
Set is max # of vertices. Largest independent

Graph 1: $\{1, 4, 6, 8, 9\}$ → 5 vertices

Graph 2: $\{A, C, G, I, M, S, V, X, Y\}$ → 9 vertices

Long explanation/Reply: No wait!

Graph 1

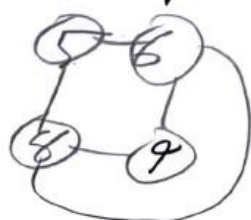


A) $\{4, 1, 7, 9\}$
maybe?

Graph B: Largest independent set = 10 vertices

$\{A, C, E, I, K, M, O, Q, S, W\}$

Graph A: clique = 4



b) Largest independent set

$\{1, 6, 7, 8\}$ $\{2, 9, 7\}$ $\{4, 2, 9\}$ $\{1, 9, 8\}$

7.5 Is the following formula satisfiable?

$$(x \vee y) \wedge (x \vee \bar{y}) \wedge (\bar{x} \vee y) \wedge (\bar{x} \vee \bar{y})$$

SAT membership 7.5

Is it satisfiable?

OR AND/8

$$(x \vee y) \wedge (x \vee \bar{y}) \wedge (\bar{x} \vee y) \wedge (\bar{x} \vee \bar{y})$$

Pretty sure No! Let's say 1 = True 0 = False make

AND

$$(0 \& 0) = F \quad 0 \& 1 = 0 \quad 1 \& 0 = 0 \rightarrow F \quad 1 \& 1 = 1 \rightarrow \text{True}$$

$$(1 \& 0) \& (1 \& 1) \& (0 \& 1) \& (0 \& 0) \text{ Nope!}$$

Now flip symbol

$$0 \vee 0 = 0 \quad 0 \vee 1 = 1 \quad 1 \vee 0 = 1 \quad 1 \vee 1 = 1$$

make $x=1, y=0$ ~~wait no since~~

$$(1 \vee 0) \& (1 \vee 0) \& (\bar{1} \vee 0) \& (\bar{1} \vee 0) \& (\bar{1} \vee \bar{0})$$

$$(1) \& (1) \& (0) \& (0) \& (1) = \text{X Nope}$$

Thus cannot satisfy.

Done

Did that in my head
 $x=0, y=1$
 accidentally

~~$$(1 \vee 0) \& (1 \vee 0) \& (\bar{1} \vee 0) \& (\bar{1} \vee 0) \& (\bar{1} \vee \bar{0})$$~~

0.1 Examine the following formal descriptions of sets so that you understand which members they contain. Write a short informal English description of each set.

- a. $\{1, 3, 5, 7, \dots\}$
- b. $\{\dots, -4, -2, 0, 2, 4, \dots\}$
- c. $\{n \mid n = 2m \text{ for some } m \text{ in } \mathcal{N}\}$
- d. $\{n \mid n = 2m \text{ for some } m \text{ in } \mathcal{N}, \text{ and } n = 3k \text{ for some } k \text{ in } \mathcal{N}\}$
- e. $\{w \mid w \text{ is a string of 0s and 1s and } w \text{ equals the reverse of } w\}$
- f. $\{n \mid n \text{ is an integer and } n = n + 1\}$

DSR: 0.1 - 0.5 Prove by induct 0.6, 0.8, 0.9, 0.10, 0.11

0.1:

0.1A $\{1, 3, 5, 7, \dots\}$ $\nearrow$ Natural number
all odd $\mathbb{N}$'s

0.1B $\{ \dots, -4, -2, 0, 2, 4, \dots \}$

The $\mathbb{N}$ set of all even integers

0.1C $\{n \in \mathbb{N} \mid n = 2m \text{ for some } m \in \mathbb{N}\}$

• All even $\mathbb{N}$'s

• $\mathbb{N}$'s that can be divided by 2.

0.1D $\{n \in \mathbb{N} \mid n = 2m \text{ for some } m \in \mathbb{N}, \& n = 3k \text{ for some } k \in \mathbb{N}\}$

the set of odd.

Potentially... all of the $\mathbb{N}$'s can be divided by either 2 or 3.

• All $\mathbb{N}$ can be $\times$ by 6.

0.1E $\{w \mid w \text{ is a string of 0's \& 1's \& } w = \text{the reverse of } w^R\}$

It's a thing defined as palindrom. All what all of the string is. It is the set of string that only contains 0's & 1's.

0.1F $\{n \in \mathbb{N} \mid n \text{ is an integer \& } n = n+1\}$

Pretty weird I think it would be $\emptyset$ because no $\mathbb{N}$ can be $n+1$ or n . However it's kind of saying the set of all $\mathbb{N}$'s or ints that are equal to $n+1$ to the $\mathbb{N}$.

0.2 Write formal descriptions of the following sets.

- a. The set containing the numbers 1, 10, and 100
- b. The set containing all integers that are greater than 5
- c. The set containing all natural numbers that are less than 5
- d. The set containing the string aba
- e. The set containing the empty string
- f. The set containing nothing at all

0.2 The set containing $\rightarrow$ In each section

0.2A The #s 1, 10, & 100

$\{1, 10, 100\}$ $\parallel \{n \mid n = 10^m \text{ \& for sum } m \in \{0, 1, 2\}\}$

$\uparrow$
on b/c it has the # of 10 in series.

0.2B All integers that are greater than 5

$\{n \mid n \text{ is 1 or more ints \& } n \geq 5\}$

b/c is of all of the ints they are ≥ 5

0.2C ~~The~~ All $\mathbb{N}$ #'s that are less than 5

$\{n \mid n \in \mathbb{N}, n < 5\}$

< Not gonna do b/c $\rightarrow$ kind of obvious / common sense

0.2D The string abba
 $\{n \mid n \text{ has } \dots \{a b a\}\}$

aba = string so $\{ \}$

0.2E The $\{\}$ empty ~~set~~ string

$\{ \{ \}$ empty $\} = \{ \{n \mid n \in \{ \} \}$

0.2F Containing nothing at all.

No

$\rightarrow$ means Null or $\emptyset$

0.3 Let A be the set $\{x, y, z\}$ and B be the set $\{x, y\}$.

- a. Is A a subset of B ?
- b. Is B a subset of A ?
- c. What is $A \cup B$?
- d. What is $A \cap B$?
- e. What is $A \times B$?
- f. What is the power set of B ?

0.3A Let A be the set $\{x, y, z\}$ & B be the set $\{x, y\}$ for everything
 Is A a subset of B ?
 No, A has elements z & it's not in B .

0.3B Is B a subset of A ?
 yes, All/every member $= B$ is a member of A . Thus A proper subset of A is B .

0.3C What is $A \cup B$?
 $A = \{x, y, z\}$
 $B = \{x, y\}$
 Or
 $\{x, y, z\} \cup \{x, y\} \rightarrow \{x, y, z\} \rightarrow A$
 comp Does this $\supset A$

0.3D $A \cap B$?
 $\{x, y, z\} \cap \{x, y\} = \{x, y\} = B$
 And means what do both contain. thus B .

0.3E $A \times B$? $x = \text{unit values}$
 $\{(x, x), (x, y), (x, x, y, y), (z, x), (z, y)\}$
 Or
 $\{(x, x), (x, y), (x, z), (x, y), (y, y), (x, z)\}$

0.3F What is the power set of B ? Not necessarily Lob
 $P_B = \{\emptyset, \{x\}, \{y\}, \{x, y\}\}$ $\emptyset = \text{Null}$
 from from B but separate.

x	y	z
x	y	$\emptyset$

0.4 If A has a elements and B has b elements, how many elements are in $A \times B$? Explain your answer.

0.4
 $A \times B$ is doing cartesian product $A(x_a)$ pair it
 w/ $B(y_1, y_2, y_3, \dots)$ & you get those sets.
 Also discrete # of elements $\rightarrow$ Thus Final $+1$ $\leftarrow$ how many
 combos.
 $a \times b$

 $\sum_{i=1}^n (x_i, y_1), (x_i, y_2), (x_i, y_3), \dots$
 $n \neq 1$
 do
 $\{ (x_1, y_1), (x_1, y_2), (x_1, y_3), (x_1, y_4), \dots \}$
 Repeat when used all combos of y . Go to the next x
 then = done.
 $A \times B$ will get $a \times b$ # of elements from it.

0.5 If C is a set with c elements, how many elements are in the power set of C ? Explain your answer.

Let C be a set w/ C elements. The # of elements in power set $P(C)$ is 2^C .

C has 0 elements ($C=0$), then $P(C)$ is the set containing only the $\emptyset$. Thus, $|P(C)| = 1 = 2^0$

Assume the statement holds for $C=k$, i.e., $|P(C)| = 2^k$ for a set C w/ k elements.

Consider a set C w/ $k+1$ elements. Can form $P(C)$ by taking the set of a set w/ k ele. & + 1 new ele. to each subset. $2 \times$ # of subsets

$$|P(C)| = 2 \times 2^k = 2^{k+1}$$

Thus by induction power set of a set w/ C ele has 2^C ele.

0.6 Let X be the set $\{1, 2, 3, 4, 5\}$ and Y be the set $\{6, 7, 8, 9, 10\}$. The unary function $f: X \rightarrow Y$ and the binary function $g: X \times Y \rightarrow Y$ are described in the following tables.

n	$f(n)$
1	6
2	7
3	6
4	7
5	6

g	6	7	8	9	10
1	10	10	10	10	10
2	7	8	9	10	6
3	7	7	8	8	9
4	9	8	7	6	10
5	6	6	6	6	6

- What is the value of $f(2)$?
- What are the range and domain of f ?
- What is the value of $g(2, 10)$?
- What are the range and domain of g ?
- What is the value of $g(4, f(4))$?

2.6

Let x be the set $\{1, 2, 3, 4, 5\}$ & $y \in \{6, 7, 8, 9, 10\}$

$f: x \rightarrow y$ binary funct $xx \rightarrow y$

n	$f(n)$	n	6	7	8	9	10
1	6	1	10				
2	7	2	7	8	9	10	6
3	6	3	7	7	8	8	9
4	7	4	9	8	7	6	10
5	6	5	6				

Q6A) the funct $f(n)$ if... $n=2 \Rightarrow$ the $q(2) =$ what is across
 $n=2$ it is $\rightarrow f(2)=7$

0.68)

Range = dep v domain = indep. w/ or

a cause the set of All input s.

$$f(n) = y = \text{Range} = R = \{6, 7\}$$

$$n = x = \text{Domain} = D = \{1, 2, 3, 4, 5\} \text{ or } q \text{ is also}$$

06C) Go to Row that's = to 2. & then Col. that's = 108
the # on the box = 6.

$$q(2, 10) = 6$$

0.6D) Range = $\{6, 7, 8, 9, 10\}$

Domain = $\{1, 2, 3, 4, 5\}$

Side Note: I'm not gonna do, between too much work

$x \leftarrow$ them

$(1, 6) (1, 7) (1, 8) (1, 9) (1, 10) (2, 6) (2, 7) (2, 8) (2, 9) (2, 10)$

$(3, 6) (3, 7) (3, 8) (3, 9) (3, 10) (4, 6) (4, 7) (4, 8) (4, 9) (4, 10)$

$(5, 6) (5, 7) (5, 8) (5, 9) (5, 10)$

06E) $q(4, f(4))$

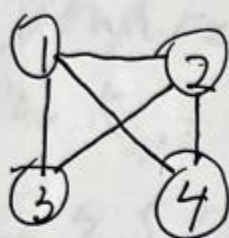
Find coordinate $f(4) = 7$ call it. $f(4)$

$n=4$ & it gives us to 7

$$f(4) = 7 \quad (4, f(4)) = q(4, 7) \rightarrow 8$$

0.8 Consider the undirected graph $G = (V, E)$ where V , the set of nodes, is $\{1, 2, 3, 4\}$ and E , the set of edges, is $\{\{1, 2\}, \{2, 3\}, \{1, 3\}, \{2, 4\}, \{1, 4\}\}$. Draw the graph G . What are the degrees of each node? Indicate a path from node 3 to node 4 on your drawing of G .

0.8. $G = (V, E)$ V , the set of Node $\{1, 2, 3, 4\}$
 $E = \text{edges } \{1, 2\}, \{2, 3\}, \{1, 3\}, \{2, 4\}, \{1, 4\}$. Draw the graph.
 What are the degrees of each Node? Indicate a path from node 3 to node 4.
 $N = \text{Node}$
 1. Draw Node $1 \rightarrow 4$



$N_1 : 2, 3, 4 \rightarrow \text{Deg } 3$

$N_2 : 1, 3, 4 \rightarrow \text{Deg } 3$

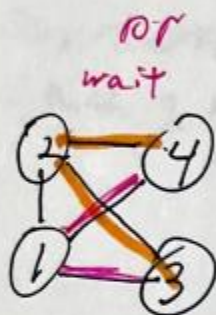
$N_3 : 1, 2 \rightarrow \text{Deg } = 2$

$N_4 : 1, 2 \rightarrow \text{Deg } = 2$

Path $N_3 \rightarrow N_4$

Path $3 \rightarrow 1 \rightarrow 4$ is the shortest

$\{1, 2\}, \{2, 3\}, \{2, 4\}, \{1, 4\}$



N	1	2	3	4
D	3	3	2	2

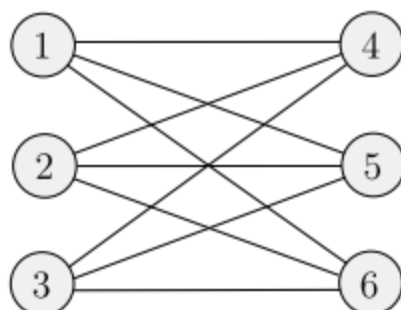
What is shortest from $3 \rightarrow 4 \Rightarrow 3 \rightarrow 1$, then $1 \rightarrow 4$
 $= 2$. Others are

Longer. & it's unweighted.

$3 \rightarrow 1 \rightarrow 4$

$3 \rightarrow 2 \rightarrow 4$

0.9 Write a formal description of the following graph.



Q9.

Set of Nodes (verts) $V = \{1, 2, 3, 4, 5, 6\}$

Set of Edges $E = \{(1, 4), (1, 5), (1, 6), (2, 4), (2, 5), (2, 6), (3, 4), (3, 5), (3, 6)\}$

This graph represents a complete bipartite graph $K_{3,3}$. Where the 2 sets of vert are $\{1, 2, 3\}$ & $\{4, 5, 6\}$ with each node in the 1st set connected to every node in the second set. each going to 3 Nodes.

PROBLEMS

0.10 Find the error in the following proof that $2 = 1$.

Consider the equation $a = b$. Multiply both sides by a to obtain $a^2 = ab$. Subtract b^2 from both sides to get $a^2 - b^2 = ab - b^2$. Now factor each side, $(a + b)(a - b) = b(a - b)$, and divide each side by $(a - b)$ to get $a + b = b$. Finally, let a and b equal 1, which shows that $2 = 1$.

10

$$2=1$$

Given: $A=b$

- x both sides by a to get $a^2 - b^2 = ab - b^2$
- Subtract b^2 from both sides to get $a^2 - b^2 = ab - b^2$
- factor $(a+b)(a-b) = b(a-b)$

Crit Error: / both sides by $(a-b)$ since $a=b, a-b=0$
 so / by 0 is undef.

The proof is invalid because / by 0 is not allowed
 & this step leads to an incorrect conclusion that
 $2=1$.

~~$$(a+b)$$~~

$$a=b$$

$$a^2 = ab - b^2$$

$$-b^2 + a^2 = -b^2 + ab$$

$$(a+b)(a-b) = b(a-b)$$

$$a+b = b \quad \text{if } a=b=1$$

$$1+1=2 \rightarrow 2=1$$

but / $(a-b)$ = wrong b/c not def.

0.11 Let $S(n) = 1 + 2 + \dots + n$ be the sum of the first n natural numbers and let $C(n) = 1^3 + 2^3 + \dots + n^3$ be the sum of the first n cubes. Prove the following equalities by induction on n , to arrive at the curious conclusion that $C(n) = S^2(n)$ for every n .

a. $S(n) = \frac{1}{2}n(n+1)$.

0.11 a

$$S(n) = \frac{1}{2}n(n+1)$$

$$n=1$$

$$S(1) = 1 = \frac{1}{2}(1)(1+1) = \frac{1}{2} \times 2 = 1$$

Assume form holds for $n=k$. i.e., $S(k) = \frac{1}{2}k(k+1)$

$$n=k+1$$

$$S(k+1) = S(k) + (k+1)$$

Substitute inductive hypothesis

$$S(k+1) = \frac{1}{2}k(k+1) + (k+1)$$

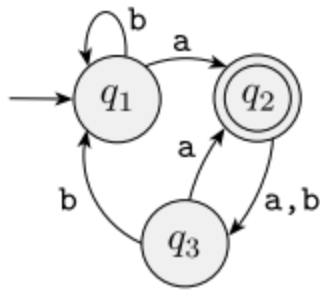
Factor out $(k+1)$

$$S(k+1) = \frac{1}{2}(k+1)(k+2)$$

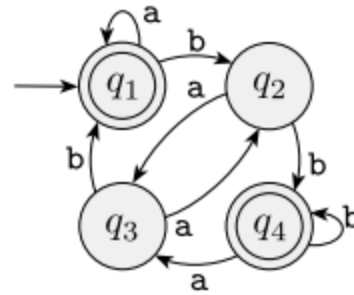
matches

by induction $S(n) = \frac{1}{2}n(n+1)$ is true for all n .

^A**1.1** The following are the state diagrams of two DFAs, M_1 and M_2 . Answer the following questions about each of these machines.



M_1



M_2

- a. What is the start state?
- b. What is the set of accept states?
- c. What sequence of states does the machine go through on input aabb?
- d. Does the machine accept the string aabb?
- e. Does the machine accept the string ϵ ?

^A**1.2** Give the formal description of the machines M_1 and M_2 pictured in Exercise 1.1.

Q.1 (m2 only)

1. q_1 = start state
2. Accepted states $\{q_1, q_2, q_4\}$
3. Sequence of states for input = "aabb"

q_1
input a → move to q_2 .
a → stay at q_2
b → Move to q_4
b → stay at q_4

seq: $q_1 \rightarrow q_2 \rightarrow q_2 \rightarrow q_4 \rightarrow q_4$

4. yes the machine accepts "aabb" b/c it ends in the accept state q_4

5. yes the machine accepts ϵ b/c the start state q_1 is also an accept state

1.2 cm2 only)

1. Formal Description of DFA M_2 :

Set of states $Q = \{q_1, q_2, q_3, q_4\}$, ϵ, δ, q_1, F

Alphabet $\Sigma: \{a, b\}$

Start State: q_1

Set of Accept states $F: \{q_1, q_2, q_4\}$

Transition function δ :

$\delta(q_1, a)$	q_2
(q_1, b)	q_3
(q_2, a)	q_2
(q_2, b)	q_4
(q_3, a)	q_4
(q_3, b)	q_1
(q_4, a)	q_3
(q_4, b)	q_4

1.6 Give state diagrams of DFAs recognizing the following languages. In all parts, the alphabet is $\{0,1\}$.

a. $\{w \mid w \text{ begins with a 1 and ends with a 0}\}$

b. $\{w \mid w \text{ contains at least three 1s}\}$

c. $\{w \mid w \text{ contains the substring 0101 (i.e., } w = x0101y \text{ for some } x \text{ and } y)\}$

d. $\{w \mid w \text{ has length at least 3 and its third symbol is a 0}\}$

e. $\{w \mid w \text{ starts with 0 and has length at least 3}\}$

f. $\{w \mid w \text{ contains the substring 010}\}$

g. $\{w \mid \text{the length of } w \text{ is at most 5}\}$

h. $\{w \mid w \text{ is a string except 11 and 111}\}$

i. $\{w \mid w \text{ contains exactly one 1}\}$

j. $\{w \mid w \text{ contains at least two 0s and at most one 1}\}$

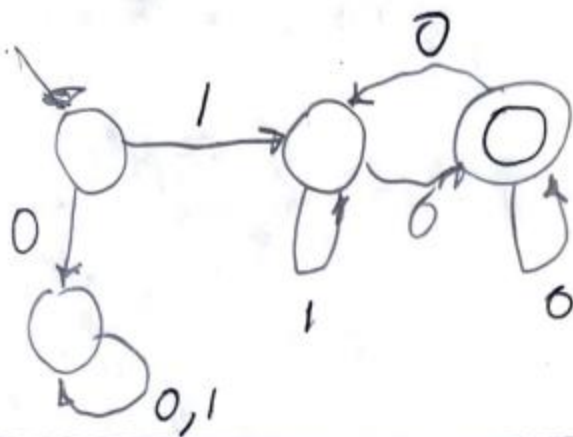
k. $\{\epsilon, 0\}$

l. $\{w \mid w \text{ contains an even number of 0s and an even number of 1s}\}$

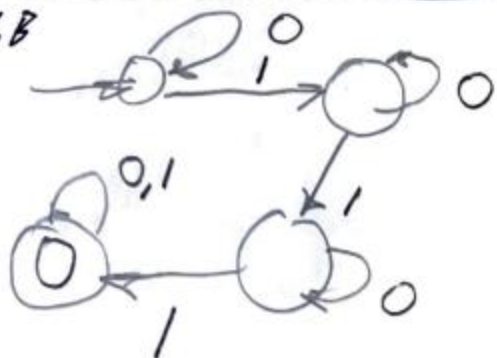
m. The empty set

n. All strings except the empty string

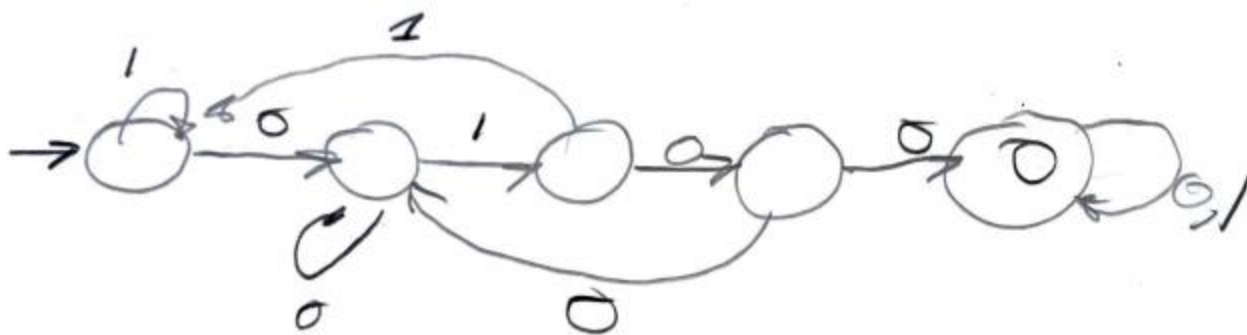
PSA Construction 1.6 $8 \neq 4$
1.6A



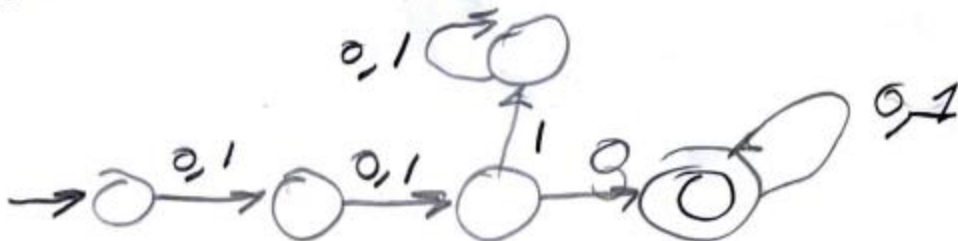
1.6B

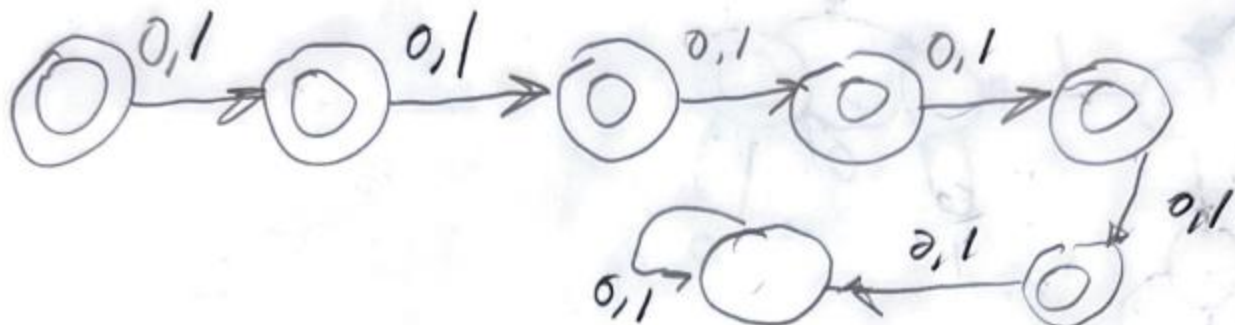


1.6C

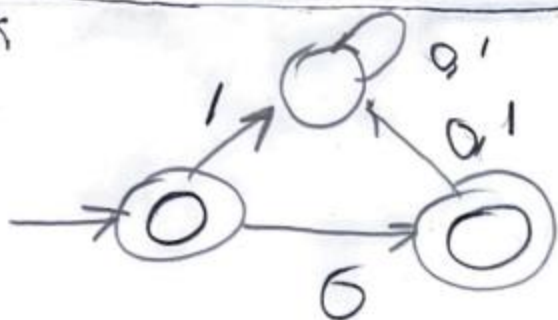


1.6D

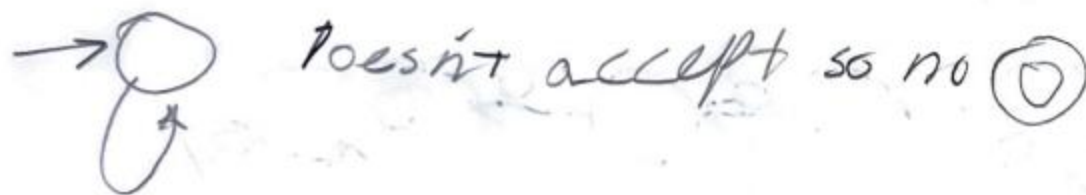




1.6 K



1.6 M



1.6 N

